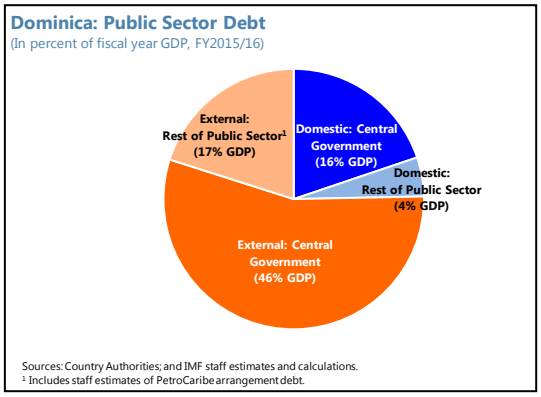
Annex I. Debt Sustainability Analysis[1]

Risk of external debt distress:	*High*

Dominica continues to be assessed at high risk of debt distress. Setting public and external debt on a declining path requires reaching the fiscal consolidation target of over 6 percent of GDP by FY2020/21 committed by the authorities in the RCF disbursement. In addition, given lower projected grant flows, reaching the regional debt target of 60 percent of GDP by 2030 calls for the use of accumulated citizenship-by-investment (CBI) deposits for debt reduction. Main risks to the debt sustainability outlook include the failure to advance with the fiscal consolidation; a sudden stop in CBI revenues; and further weakening of the prospects for donor grants. Another large natural disaster would find the government with no buffers.

A. Background

1. Dominica's overall public debt has been increasing steadily since 2010. As of end-FY 2015/16, the stock of public sector debt (central government and rest of the public sector[2], henceforth public debt) is estimated to be around 83 percent of GDP. Over ¾ of this stock is external debt, owed largely to multilateral and bilateral creditors and the remaining is held domestically, mostly by commercial banks and other financial institutions. The debt of the rest of public sector (state owned enterprises and Petrocaribe debts) is about 20 percent of GDP.



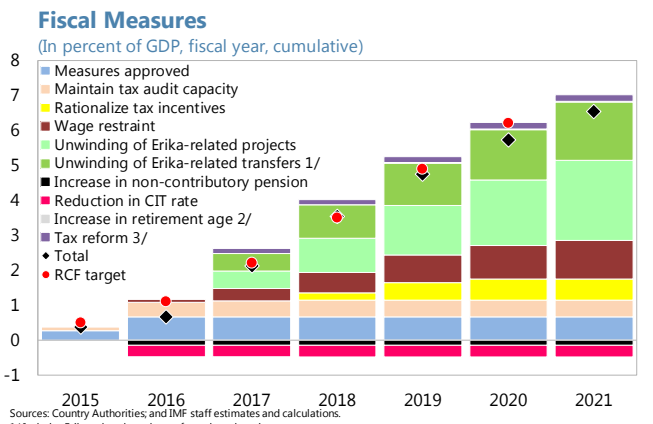
2. After being hit by the Tropical Storm Erika in August 2015, the Fund approved an RCF disbursement. The storm produced damages estimated at 96 percent of GDP, causing extensive impairment to public and private infrastructure. The RCF disbursement provided short-term financing relief, and a medium-term fiscal consolidation plan including measures targeting a cumulative adjustment of over 6 percent of GDP over a five-year period, as needed to reach the regional debt target of 60 percent of GDP by 2030. The government is pursuing a gradual fiscal consolidation over the medium-term, including comprehensive reforms to increase tax revenues and

[1] This DSA has been cleared by Mr. Bob Traa (IMF) and Ms. Paloma Anos-Casero (World Bank).

[2] Debt of the rest of the public sector also includes staff's estimate of Petrocaribe arrangement. The estimation is based on the Petrocaribe loan agreement. It should be noted, however, that Petrocaribe debt service payments have been cancelled or rescheduled in the last three years, a pattern that may continue into the coming years. However, this DSA includes these obligations for prudence given the uncertainty about the continuation of this practice.

broaden the tax base, and contain current expenditures. The initial fiscal measures discussed in the RCF disbursement have been adopted, and key technical assistance to advance on the second-generation fiscal reforms has been delivered or is expected before the end of FY2016/17. The government has also passed additional measures to those discussed in the RCF disbursement to further strengthen the fiscal position. This debt sustainability outlook includes an annualized cost of rehabilitation and reconstruction after natural disasters of 1.5 percent of GDP per year, providing a degree of resilience to the fiscal consolidation strategy.

3. If the fiscal consolidation plan is fully implemented, the debt outlook would improve significantly, setting public debt on a declining path. The authorities have been taking measures that until now have been broadly in line with the targets, and are working to implement additional measures to reach the consolidation target. The debt sustainability analysis is based on an active policies scenario in which the government reaches the fiscal consolidation targets in the RCF disbursement.



Sources: Country Authorities; and IMF staff estimates and calculations.
1/ Includes Erika-related purchase of goods and services.
2/ Includes increase in employer contributions.
3/ Review of healthcare fees to improve cost recovery and introduction of a presumptive tax. FAD TA formally requested and PAHO technical support being provided.

4. At the government's request, Paris Club creditors considered a debt relief program in 2016, but ultimately explained that no debt treatment would be granted without a Fund-supported program. In light of the high debt burden, the large fiscal costs of the reconstruction of public infrastructure after the tropical storm, and its protracted effects on output, the authorities have formally requested debt relief from the Paris Club as part of their fiscal consolidation strategy. The authorities are also in conversations with the government of Venezuela which could reduce Petrocaribe obligations within the context of an early repayment at a discount.

B. Underlying Assumptions

5. Macroeconomic assumptions underlying the DSA are summarized in Box 1. The main differences relative to the DSA in the 2016 Staff Report, which contribute to higher PV of public sector external debt in 2016 are as follows:

- The potential growth rate has been revised down based on analytical inputs and recent trends in investment, employment, and productivity.
- The long-term fiscal outlook has also deteriorated mainly as a result of further downward revisions in the projection of grants.
- Observed CBI revenues have been higher than previously estimated, but these flows are unpredictable and subject to the risk of a sudden stop. The increased dependence on these inflows for government and external financing represents an increase in overall risk.

- Grants have been revised to less than half of the level in previous projections, based on recent observed performance and international trends. As noted in the 2016 Article IV Staff Report, external and public debt sustainability are highly sensitive to these flows.
- Some revisions and corrections to historical debt data and the corresponding future payment obligations which affect the PV calculations.

Box A1. Underlying Assumptions (2016-36)

The active scenario assumes fiscal consolidation of over 6 percent of GDP through 2016-22. During this period, public investment is projected to increase underpinned by the reconstruction of public infrastructure after Tropical Storm Erika. This results in an acceleration of growth in the near term, which gradually declines towards a potential growth rate of 1.5 percent. Some large scale investment projects under execution contribute to higher growth rates in the near term. During that period, the fiscal and external imbalances deteriorate before improving into the medium term on the back of fiscal consolidation and the unwinding of investment for reconstruction.

- **Real GDP.** Real GDP growth increases to over 3 percent in the near term and gradually declines towards the potential rate of 1.5 percent by 2022.
- **Inflation.** Inflation is projected to remain near 2 percent in line with international inflation, given the quasi-currency board arrangement of Eastern Caribbean dollar.
- **Balance of Payments.** Current account deficit remains high throughout the projection period financed mainly with FDI and official debt flows.

C. External DSA

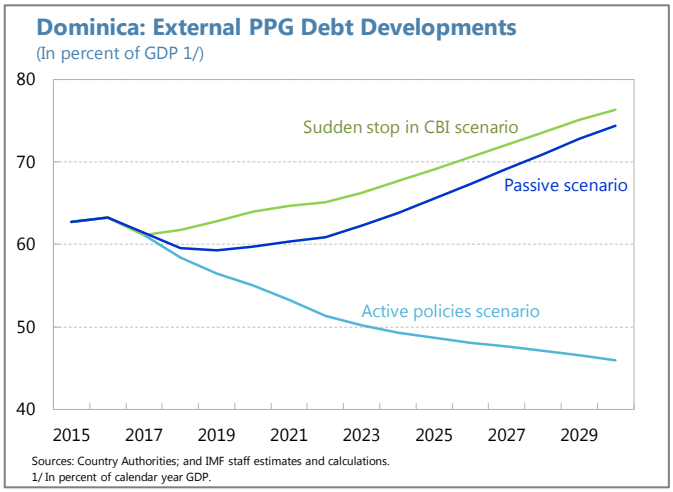
6. The rating assigned to PPG external debt is "high risk of debt distress". In the active scenario, and without further shocks, the PV of debt-to-GDP ratio is breached (see Figure A1). Other debt stock and debt service measures, however, remain below the policy-dependent indicative thresholds during the entire forecast period in the active scenario.

7. Moreover, under the historical and various alternative scenarios, external PPG debt dynamics become unsustainable for most debt indicators. Several risk thresholds are breached in the historical and shock scenarios. Under the historical scenario – which assumes non-interest current account balance, FDI, real GDP growth, and the GDP deflator remain at their 10-year historical average – the PV of debt-to-GDP threshold is breached for the entire period, and the PV of public external debt takes an increasing trajectory. The PV of debt-to-exports, debt service-to-exports, and debt service-to-revenue ratios also breach the thresholds under the historical scenario. In addition, under a combination shock[3], the PV of debt-to-GDP ratio returns to below threshold levels only at the end of the forecast horizon – by 2032, while the debt service-to-revenue ratio

[3] Combination of growth, exports, GDP deflator, and non-debt creating flows shocks using one-half standard deviation in 2017-2018.

threshold is breached briefly under a one-time depreciation shock. For most other indicators (PV of debt-to-exports, debt-to-revenue and most of the debt service ratios) the PV of external debt remains below thresholds under the temporary shock scenarios.

8. The results of the customized scenarios, designed to pinpoint sources of high risk, further justify the classification of "high risk of external debt distress". To highlight some of the main potential risks not captured in the standard stress tests, two additional customized scenarios are included in the analysis (text chart). First, assuming that the remaining fiscal consolidation is not passed because of capacity constraints or social resistance, public debt would take an increasing trajectory (passive scenario). Second, a scenario of sudden stop in the CBI revenues in FY2018/19 in the active policies scenario highlights the downside risk of the increasing reliance on unpredictable and volatile CBI revenues. The simulations indicate that materialization of these risks would result in a permanent breach of the PV of debt-to-GDP ratio threshold with an upward trajectory, rendering debt dynamics unsustainable as shown in Figure A1.



9. Based on the threshold breaches in active, shock, and customized scenarios, Dominica's external PPG debt is rated as "high risk of debt distress". Given the challenges associated with fiscal consolidation in the wake of storm Erika, the customized scenarios, which also simulate an upward trending PV of debt-to-GDP, are attached significant weight in the risk assessment as they attempt to capture important vulnerabilities in the economy overlooked by the standardized tests. As the planned fiscal consolidation takes hold, further increasing confidence in the active policies scenario, an upgrade of the risk rating could be considered in a future assessment.

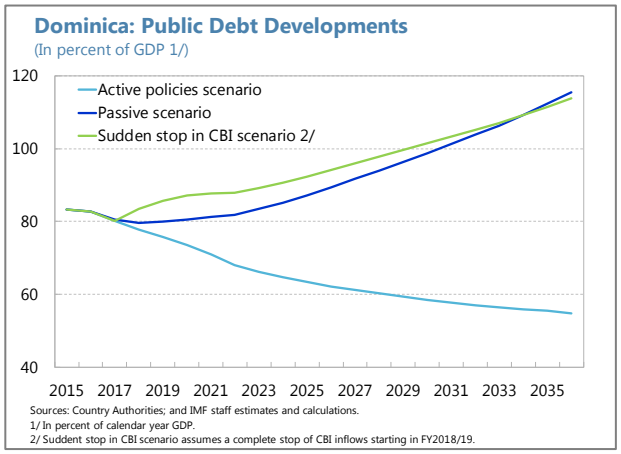
D. Public DSA

10. In an active policies scenario public debt is assessed to be sustainable. Under the active policies scenario – which assumes that the government completes its fiscal consolidation plan and meets the targets discussed in the RCF disbursement – the PV of public sector debt was estimated at 73 percent of GDP in 2016 and is expected to remain on a declining trajectory below the public debt threshold during the entire projection period. The PV of the debt-to-revenue ratio also declines, while debt service-to-revenue ratio increases over the forecast horizon (see Figure A2).

11. Simulations with alternative shock scenarios point to elevated but below the benchmark PPG debt levels. The most extreme shock (growth shock) in which real GDP is set to

historical average minus one standard deviation in 2017-18, leads to an unsustainable and increasing PV of debt-to-GDP ratio. Alternative scenarios, which assume that primary balance-to-GDP ratio and real GDP growth are set to their historical averages (historical scenario), and primary balance-to-GDP ratio is set to its value in 2016 (fixed primary balance scenario), generate stable debt trajectories. This implies no margin for unexpected negative shocks under the assumptions in these scenarios.

12. Under the customized scenarios total PPG debt takes on an increasing trajectory. Under the two customized scenarios, which are included to highlight some of the main risks not captured in the standard stress tests, PV of debt-to-GDP, debt-to-revenue, and debt-service to revenue ratios continue to increase throughout the forecast period, breaching the public debt benchmark in 2018.

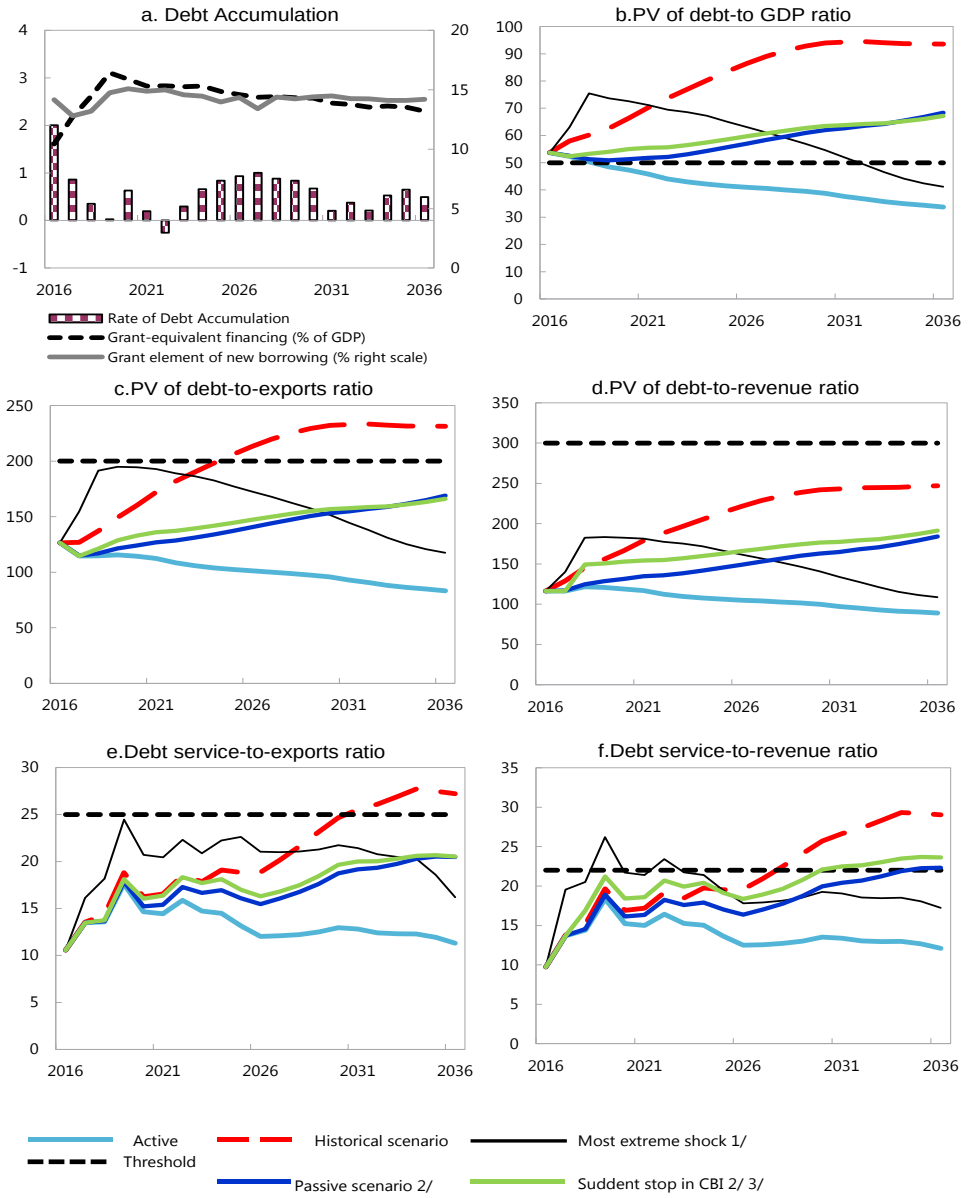


E. Conclusion

13. The large debt level and the significant risks to the debt sustainability outlook merit the classification of Dominica as "high risk of external debt distress". This assessment is based on debt dynamics presented in the active policies scenario and alternative scenarios, as well as the risks reflected in the customized scenarios and the significant risks explained above, which carry significant weight in this assessment.

14. The authorities agreed with Staff that Dominica remains at high risk of debt distress. They indicated that efforts to reduce debt levels must continue. They agreed that debt dynamics could become unsustainable if the fiscal consolidation package committed under the Rapid Credit Facility (RCF) was not implemented. In addition, they acknowledged the importance of grants and CBI flows to finance post-Erika capital program and for debt reduction, without which the debt thresholds would be breached. Therefore, the authorities reiterated their commitment to the fiscal consolidation targets in the RCF disbursement and agreed with Staff on the need to consider contingent fiscal measures. Finally, the authorities noted that only the portion of Petrocaribe funds, which has been approved as grants, has been utilized. Therefore, they questioned the inclusion of Petrocaribe obligations in the stock of public debt. However, Petrocaribe debt remains a gross liability of the government.

Figure A1. Dominica: Indicators of Public and Publicly Guaranteed External Debt Under Alternative Scenarios, 2016-36 (Active Scenario) [1]



Sources: Country authorities; and staff estimates and projections.

1/ The most extreme stress test is the test that yields the highest ratio on or before 2026. In figure b. it corresponds combination shock; in c. to exports shock; in d. to a combination shock; in e. to a one time depreciation shock and i figure f. to a one-time depreciation shock. Shocks are applied to active scenario.
2/ Customized scenarios.
3/ Suddent stop in CBI scenario assumes a complete stop of CBI inflows starting in FY2018/19.

Table A1. Dominica: External Debt Sustainability Framework, Active Scenario, 2013-36 1/
(In percent of GDP) [1]

	Actual			Historical 6/ Average	Standard 6/ Deviation	Projections									
	2013	2014	2015			2016	2017	2018	2019	2020	2021	2016-2021 Average	2026	2036	2022-2036 Average
External debt (nominal) 1/	**88.2**	**92.8**	**100.7**			**103.1**	**100.5**	**96.4**	**91.7**	**87.7**	**83.6**		**66.2**	**55.4**	
of which: public and publicly guaranteed (PPG)	62.8	61.4	62.7			63.2	61.1	58.4	56.5	55.0	53.3		48.1	41.3	
Change in external debt	11.9	4.6	7.9			2.4	-2.7	-4.0	-4.7	-4.0	-4.1		-2.0	-0.9	
Identified net debt-creating flows	1.7	-0.1	3.4			1.8	-1.7	0.6	1.4	0.9	0.5		2.0	1.8	
Non-interest current account deficit	**7.0**	**6.4**	**4.5**	**12.8**	**6.5**	**8.6**	**8.9**	**13.3**	**12.7**	**10.0**	**7.1**		**7.6**	**7.3**	7.6
Deficit in balance of goods and services	9.9	10.7	9.0			11.9	12.4	16.8	16.2	13.4	10.4		10.7	9.7	
Exports	38.1	41.1	40.5			42.4	45.7	43.9	41.9	41.3	40.8		40.5	40.5	
Imports	47.9	51.8	49.5			54.3	58.1	60.6	58.1	54.8	51.2		51.2	50.2	
Net current transfers (negative = inflow)	-4.0	-4.8	-5.5	-4.3	0.7	-4.6	-4.6	-4.6	-4.6	-4.6	-4.6		-4.6	-4.5	-4.6
of which: official	-1.1	-1.0	-1.1			-0.6	-0.6	-0.6	-0.6	-0.6	-0.6		-0.6	-0.5	
Other current account flows (negative = net inflow)	1.1	0.6	1.0			1.4	1.1	1.1	1.1	1.2	1.3		1.5	2.1	
Net FDI (negative = inflow)	**-4.6**	**-6.3**	**-6.6**	**-8.2**	**2.5**	**-9.0**	**-10.3**	**-12.7**	**-12.6**	**-10.5**	**-8.3**		**-7.6**	**-7.4**	-7.6
Endogenous debt dynamics 2/	**-0.6**	**-0.2**	**5.4**			**2.2**	**-0.3**	**0.1**	**1.3**	**1.4**	**1.7**		**2.0**	**1.8**	
Contribution from nominal interest rate	2.7	3.1	3.4			3.1	3.2	3.3	3.3	3.2	3.2		3.0	2.6	
Contribution from real GDP growth	-0.6	-3.6	1.7			-1.0	-3.6	-3.2	-2.0	-1.8	-1.5		-1.0	-0.8	
Contribution from price and exchange rate changes	-2.8	0.3	0.3			...	...	...	...	...	...		...	...	
Residual (3-4) 3/	**10.2**	**4.7**	**4.5**			**0.6**	**-1.0**	**-4.7**	**-6.1**	**-4.9**	**-4.7**		**-4.0**	**-2.7**	
of which: Capital transfers						2.4	2.0	2.6	2.9	3.0	2.9		2.8	2.4	
of which: Commercial Banks and other private flows						2.4	0.3	0.1	-1.3	-1.8	-2.5		-3.9	-3.8	
of which: exceptional financing	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
PV of external debt 4/	...	...	90.2			93.6	91.7	88.3	83.8	80.0	76.1		59.1	47.8	
In percent of exports	...	...	222.8			220.5	200.5	201.4	199.7	193.4	186.5		145.9	118.2	
PV of PPG external debt	**...**	**...**	**52.2**			**53.7**	**52.4**	**50.3**	**48.5**	**47.3**	**45.8**		**41.0**	**33.7**	
In percent of exports	**...**	**...**	**129.0**			**126.5**	**114.5**	**114.8**	**115.6**	**114.4**	**112.3**		**101.2**	**83.3**	
In percent of government revenues	**...**	**...**	**128.7**			**116.4**	**116.4**	**121.6**	**120.7**	**119.0**	**116.8**		**105.1**	**89.0**	
Debt service-to-exports ratio (in percent)	**14.6**	**14.9**	**19.4**			**15.2**	**17.2**	**17.4**	**21.5**	**18.4**	**18.3**		**15.6**	**14.4**	
PPG debt service-to-exports ratio (in percent)	**9.6**	**10.8**	**14.7**			**10.6**	**13.5**	**13.6**	**17.6**	**14.7**	**14.4**		**12.0**	**11.3**	
PPG debt service-to-revenue ratio (in percent)	**10.1**	**13.1**	**14.6**			**9.7**	**13.7**	**14.4**	**18.4**	**15.2**	**15.0**		**12.5**	**12.1**	
Total gross financing need (Billions of U.S. dollars)	0.0	0.0	0.0			0.0	0.0	0.0	0.1	0.0	0.0		0.0	0.1	
Non-interest current account deficit that stabilizes debt ratio	-5.0	1.8	-3.3			6.2	11.6	17.3	17.4	14.1	11.2		9.6	8.2	
Key macroeconomic assumptions															
Real GDP growth (in percent)	0.8	4.2	-1.8	2.0	3.3	1.0	3.6	3.3	2.2	2.1	1.7	2.3	1.5	1.5	1.5
GDP deflator in US dollar terms (change in percent)	3.8	-0.3	-0.3	1.7	2.8	0.0	0.6	1.4	1.6	1.8	1.9	1.2	2.0	2.0	2.0
Effective interest rate (percent) 5/	3.8	3.6	3.6	4.2	0.7	3.1	3.3	3.4	3.6	3.6	3.8	3.5	4.6	4.9	4.7
Growth of exports of G&S (US dollar terms, in percent)	22.2	12.2	-3.6	5.5	11.6	5.9	12.2	0.5	-0.7	2.4	2.3	3.7	3.5	3.5	3.4
Growth of imports of G&S (US dollar terms, in percent)	-2.5	12.3	-6.4	3.2	10.6	10.8	11.5	9.3	-0.5	-2.1	-3.1	4.3	3.5	2.6	3.4
Grant element of new public sector borrowing (in percent)	...	...	...	...	...	14.2	12.8	13.2	14.8	15.1	14.9	14.1	14.4	14.2	14.3
Government revenues (excluding grants, in percent of GDP)	36.4	34.0	40.6			46.1	45.0	41.4	40.2	39.8	39.2		39.0	37.9	38.7
Aid flows (in Billions of US dollars) 7/	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
of which: Grants	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
of which: Concessional loans	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Grant-equivalent financing (in percent of GDP) 8/	...	...	...			1.6	2.2	2.6	3.1	3.0	2.8		2.7	2.3	2.6
Grant-equivalent financing (in percent of external financing) 8/	...	...	...			29.8	33.1	40.8	38.4	40.7	42.7		41.3	40.4	40.2
Memorandum items:															
Nominal GDP (Billions of US dollars)	0.5	0.5	0.5			0.5	0.5	0.6	0.6	0.6	0.6		0.8	1.1	
Nominal dollar GDP growth	4.6	3.9	-2.1			1.0	4.2	4.7	3.8	3.8	3.6	3.5	3.5	3.5	3.5
PV of PPG external debt (in Billions of US dollars)			0.3			0.3	0.3	0.3	0.3	0.3	0.3		0.3	0.4	
(PVt-PVt-1)/GDPt-1 (in percent)						2.0	0.9	0.4	0.0	0.6	0.2	0.7	0.9	0.5	0.6
Gross workers' remittances (Billions of US dollars)	...	...	...			...	...	...	...	...	...		...	...	
PV of PPG external debt (in percent of GDP + remittances)	...	...	52.2			53.7	52.4	50.3	48.5	47.3	45.8		41.0	33.7	
PV of PPG external debt (in percent of exports + remittances)	...	...	129.0			126.5	114.5	114.8	115.6	114.4	112.3		101.2	83.3	
Debt service of PPG external debt (in percent of exports + remittances)	...	...	14.7			10.6	13.5	13.6	17.6	14.7	14.4		12.0	11.3	

Sources: Country authorities; and staff estimates and projections.

1/ Includes public and private sector external debt.
2/ Derived as [r - g - ρ(1+g)]/(1+g+ρ+gρ) times previous period debt ratio, with r = nominal interest rate; g = real GDP growth rate, and ρ = growth rate of GDP deflator in U.S. dollar terms.
3/ Includes exceptional financing (i.e., changes in arrears and debt relief); changes in gross foreign assets; and valuation adjustments. For projections also includes contribution from price and exchange rate changes.
4/ Assumes that PV of private sector debt is equivalent to its face value.
5/ Current-year interest payments divided by previous period debt stock.
6/ Historical averages and standard deviations are generally derived over the past 10 years, subject to data availability.
7/ Defined as grants, concessional loans, and debt relief.
8/ Grant-equivalent financing includes grants provided directly to the government and through new borrowing (difference between the face value and the PV of new debt).

Table A2. Dominica: Sensitivity Analysis for Key Indicators of Public and Publicly Guaranteed External Debt, Active Scenario, 2016-36

	Projections							
	2016	2017	2018	2019	2020	2021	**2026**	2036
PV of debt-to GDP ratio								
Baseline	54	52	50	49	47	46	**41**	34
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2016-2036 1/	54	58	60	63	66	70	**86**	94
A2. New public sector loans on less favorable terms in 2016-2036 2	54	53	52	52	52	51	**52**	57
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2017-2018	54	55	55	53	52	50	**45**	37
B2. Export value growth at historical average minus one standard deviation in 2017-2018 3/	54	59	66	64	63	62	**55**	37
B3. US dollar GDP deflator at historical average minus one standard deviation in 2017-2018	54	53	52	51	49	48	**43**	35
B4. Net non-debt creating flows at historical average minus one standard deviation in 2017-2018 4/	54	57	63	61	60	58	**52**	37
B5. Combination of B1-B4 using one-half standard deviation shocks	54	63	75	74	73	71	**63**	41
B6. One-time 30 percent nominal depreciation relative to the baseline in 2017 5/	54	75	72	69	67	65	**58**	48
PV of debt-to-exports ratio								
Baseline	126	115	115	116	114	112	**101**	83
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2016-2036 1/	126	127	137	149	160	172	**213**	231
A2. New public sector loans on less favorable terms in 2016-2036 2	126	117	119	123	125	125	**128**	142
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2017-2018	126	115	115	116	114	112	**101**	83
B2. Export value growth at historical average minus one standard deviation in 2017-2018 3/	126	154	191	195	194	193	**172**	118
B3. US dollar GDP deflator at historical average minus one standard deviation in 2017-2018	126	115	115	116	114	112	**101**	83
B4. Net non-debt creating flows at historical average minus one standard deviation in 2017-2018 4/	126	126	143	145	145	143	**128**	90
B5. Combination of B1-B4 using one-half standard deviation shocks	126	150	181	185	184	183	**164**	107
B6. One-time 30 percent nominal depreciation relative to the baseline in 2017 5/	126	115	115	116	114	112	**101**	83
PV of debt-to-revenue ratio								
Baseline	116	116	122	121	119	117	**105**	89
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2016-2036 1/	116	129	145	156	167	179	**221**	247
A2. New public sector loans on less favorable terms in 2016-2036 2	116	119	126	129	130	130	**133**	151
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2017-2018	116	122	134	133	131	129	**116**	98
B2. Export value growth at historical average minus one standard deviation in 2017-2018 3/	116	131	159	159	158	157	**140**	98
B3. US dollar GDP deflator at historical average minus one standard deviation in 2017-2018	116	118	127	126	124	122	**109**	93
B4. Net non-debt creating flows at historical average minus one standard deviation in 2017-2018 4/	116	128	151	152	150	149	**133**	96
B5. Combination of B1-B4 using one-half standard deviation shocks	116	140	182	183	182	181	**162**	109
B6. One-time 30 percent nominal depreciation relative to the baseline in 2017 5/	116	166	173	172	170	166	**150**	127

1/ Variables include real GDP growth, growth of GDP deflator (in U.S. dollar terms), non-interest current account in percent of GDP, and non-debt creating flows.
2/ Assumes that the interest rate on new borrowing is by 2 percentage points higher than in the baseline., while grace and maturity periods are the same as in the baseline.
3/ Exports values are assumed to remain permanently at the lower level, but the current account as a share of GDP is assumed to return to its baseline level after the shock (implicitly assuming an offsetting adjustment in import levels).
4/ Includes official and private transfers and FDI.
5/ Depreciation is defined as percentage decline in dollar/local currency rate, such that it never exceeds 100 percent.

Table A2. Dominica: Sensitivity Analysis for Key Indicators of Public and Publicly Guaranteed External Debt, Active Scenario, 2016-36 (concluded)

Debt service-to-exports ratio								
Baseline	11	13	14	18	15	14	**12**	11
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2016-2036 1/	11	14	14	19	16	17	**19**	27
A2. New public sector loans on less favorable terms in 2016-2036 2	11	13	13	16	13	13	**13**	16
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2017-2018	11	13	14	18	15	14	**12**	11
B2. Export value growth at historical average minus one standard deviation in 2017-2018 3/	11	16	18	24	21	20	**21**	16
B3. US dollar GDP deflator at historical average minus one standard deviation in 2017-2018	11	13	14	18	15	14	**12**	11
B4. Net non-debt creating flows at historical average minus one standard deviation in 2017-2018 4/	11	13	14	19	16	16	**16**	12
B5. Combination of B1-B4 using one-half standard deviation shocks	11	15	16	22	19	19	**20**	15
B6. One-time 30 percent nominal depreciation relative to the baseline in 2017 5/	11	13	14	18	15	14	**12**	11
Debt service-to-revenue ratio								
Baseline	10	14	14	18	15	15	**12**	12
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2016-2036 1/	10	14	15	20	17	17	**20**	29
A2. New public sector loans on less favorable terms in 2016-2036 2	10	14	14	17	14	13	**14**	17
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2017-2018	10	14	16	20	17	17	**14**	13
B2. Export value growth at historical average minus one standard deviation in 2017-2018 3/	10	14	15	20	17	17	**17**	14
B3. US dollar GDP deflator at historical average minus one standard deviation in 2017-2018	10	14	15	19	16	16	**13**	13
B4. Net non-debt creating flows at historical average minus one standard deviation in 2017-2018 4/	10	14	15	20	17	16	**16**	13
B5. Combination of B1-B4 using one-half standard deviation shocks	10	14	16	22	19	18	**20**	15
B6. One-time 30 percent nominal depreciation relative to the baseline in 2017 5/	10	20	21	26	22	21	**18**	17
Memorandum item:								
Grant element assumed on residual financing (i.e., financing required above baseline) 6/	10	10	10	10	10	10	**10**	10

Sources: Country authorities; and staff estimates and projections.

6/ Applies to all stress scenarios except for A2 (less favorable financing) in which the terms on all new financing are as specified in footnote 2.

Table A3. Dominica: Public Sector Debt Sustainability Framework, Active Scenario, 2013-36
(In percent of GDP, unless otherwise indicated)

	Actual					Estimate					Projections				
	2013	2014	2015	Average 5/	Standard Deviation 5/	2016	2017	2018	2019	2020	2021	2016-21 Average	2026	2036	2022-36 Average
Public sector debt 1/	81.7	81.3	83.3			82.7	80.1	77.8	75.7	73.6	70.9		62.3	54.9	
of which: foreign-currency denominated	62.8	61.4	62.7			63.2	61.1	58.4	56.5	55.0	53.3		48.1	41.3	
Change in public sector debt	7.7	-0.4	2.0			-0.6	-2.6	-2.3	-2.1	-2.1	-2.6		-1.1	-0.8	
Identified debt-creating flows	-0.1	2.0	3.1			1.0	-1.5	-1.7	-0.9	-1.0	-1.3		-1.3	-0.8	
Primary deficit	1.2	2.8	-0.8	-0.3	3.9	-0.6	-1.0	-0.9	-1.1	-1.1	-1.6	-1.1	-1.7	-1.1	-1.5
Revenue and grants	39.6	36.0	42.8			47.1	46.5	43.4	42.4	42.0	41.4		41.0	39.6	
of which: grants	3.2	2.0	2.2			1.0	1.5	2.0	2.2	2.2	2.2		2.0	1.7	
Primary (noninterest) expenditure	40.8	38.8	42.0			46.5	45.5	42.5	41.4	40.9	39.8		39.3	38.6	
Automatic debt dynamics	-1.3	-0.8	4.0			1.6	-0.5	-0.8	0.1	0.1	0.3		0.4	0.4	
Contribution from interest rate/growth differential	-0.1	-2.1	3.1			0.8	-1.4	-1.3	-0.2	-0.1	0.2		0.4	0.4	
of which: contribution from average real interest rate	0.5	1.2	1.6			1.6	1.5	1.3	1.5	1.5	1.5		1.4	1.2	
of which: contribution from real GDP growth	-0.6	-3.3	1.5			-0.8	-2.9	-2.6	-1.6	-1.5	-1.3		-0.9	-0.8	
Contribution from real exchange rate depreciation	-1.1	1.3	0.9			0.8	0.9	0.5	0.3	0.1	0.0		...	...	
Other identified debt-creating flows	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	-0.1	
Privatization receipts (negative)	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	-0.1	
Recognition of implicit or contingent liabilities	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Debt relief (HIPC and other)	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Other (specify, e.g. bank recapitalization)	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Residual, including asset changes	7.7	-2.4	-1.2			-1.6	-1.1	-0.6	-1.2	-1.1	-1.3		0.2	0.0	
of which, use of deposits						0.0	-1.3	-1.4	-1.6	-1.6	-1.7		-0.1	0.0	
Other Sustainability Indicators															
PV of public sector debt	...	...	72.8			73.1	71.3	69.7	67.7	65.9	63.5		55.1	47.3	
of which: foreign-currency denominated	...	...	52.2			53.7	52.4	50.3	48.5	47.3	45.8		41.0	33.7	
of which: external	...	...	52.2			53.7	52.4	50.3	48.5	47.3	45.8		41.0	33.7	
PV of contingent liabilities (not included in public sector debt)	...	...	...			...	...	...	...	...	...		...	...	
Gross financing need 2/	7.0	8.8	6.4			6.1	9.3	7.6	10.0	8.8	8.2		7.2	6.7	
PV of public sector debt-to-revenue and grants ratio (in percent)	...	...	170.1			155.2	153.2	160.5	159.7	156.9	153.3		134.4	119.3	
PV of public sector debt-to-revenue ratio (in percent)	...	...	179.3			158.5	158.5	168.4	168.6	165.6	161.8		141.3	124.8	
of which: external 3/	...	...	128.7			116.4	116.4	121.6	120.7	119.0	116.8		105.1	89.0	
Debt service-to-revenue and grants ratio (in percent) 4/	14.7	16.5	16.8			14.4	22.1	19.6	26.0	23.6	23.7		21.9	19.6	
Debt service-to-revenue ratio (in percent) 4/	16.0	17.5	17.7			14.7	22.8	20.6	27.4	24.9	25.1		23.0	20.5	
Primary deficit that stabilizes the debt-to-GDP ratio	-6.5	3.2	-2.8			0.0	1.6	1.4	1.0	1.0	1.0		-0.6	-0.3	
Key macroeconomic and fiscal assumptions															
Real GDP growth (in percent)	0.8	4.2	-1.8	2.0	3.3	1.0	3.6	3.3	2.2	2.1	1.7	2.3	1.5	1.5	1.5
Average nominal interest rate on forex debt (in percent)	1.7	2.3	2.4	2.3	0.5	1.9	2.5	2.8	3.0	3.0	3.1	2.7	3.3	3.5	3.4
Average real interest rate on domestic debt (in percent)	2.0	4.9	3.9	2.7	2.8	6.1	6.2	5.4	5.1	4.9	4.8	5.4	4.9	4.5	4.9
Real exchange rate depreciation (in percent, + indicates depreciation)	-2.1	2.1	1.4	0.2	2.9	1.3	...	...	...	...	...	...	...	...	...
Inflation rate (GDP deflator, in percent)	3.8	-0.3	-0.3	1.7	2.8	0.0	0.6	1.4	1.6	1.8	1.9	1.2	2.0	2.0	2.0
Growth of real primary spending (deflated by GDP deflator, in percent)	-7.6	-0.7	6.2	-0.2	3.3	11.8	1.5	-3.5	-0.6	0.8	-0.9	1.5	1.5	0.8	1.3
Grant element of new external borrowing (in percent)	...	...	...	...	...	14.2	12.8	13.2	14.8	15.1	14.9	14.1	14.4	14.2	...

Sources: Country authorities; and staff estimates and projections.

1/ Public sector includes Central Government, State Owned Enterprises, and staff estimate of PetroCaribe arrangement.

2/ Gross financing need is defined as the primary deficit plus debt service plus the stock of short-term debt at the end of the last period.

3/ Revenues excluding grants.

4/ Debt service is defined as the sum of interest and amortization of medium and long-term debt.

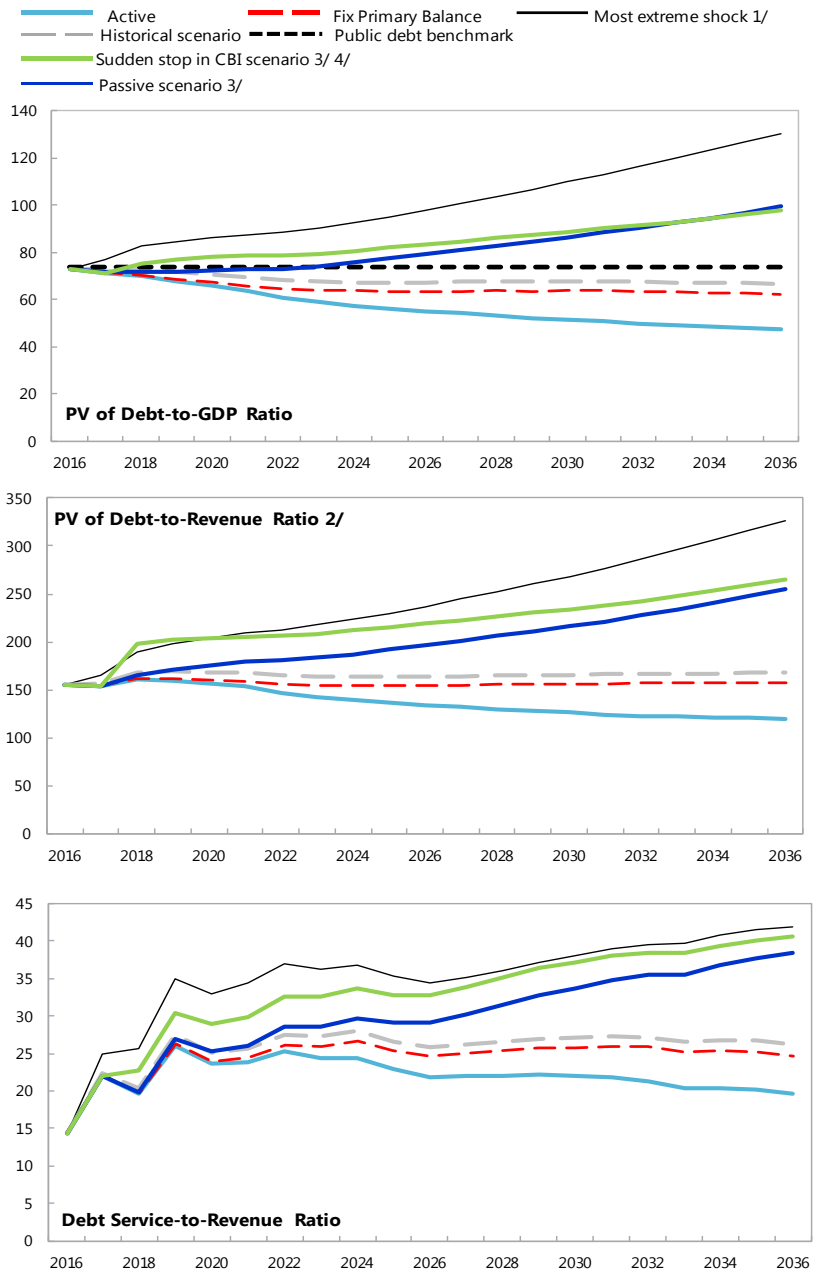
5/ Historical averages and standard deviations are generally derived over the past 10 years, subject to data availability.

Table A4. Dominica: Sensitivity Analysis for Key Indicators of Public Debt, Active Scenario, 2016-36 (In percent of GDP)

	Projections							
	2016	2017	2018	2019	2020	2021	2026	2036
PV of Debt-to-GDP Ratio								
Baseline	73	71	70	68	66	63	55	47
A. Alternative scenarios								
A1. Real GDP growth and primary balance are at historical averages	73	73	73	72	71	69	67	67
A2. Primary balance is unchanged from 2016	73	72	70	69	67	66	63	62
A3. Permanently lower GDP growth 1/	73	72	72	71	70	70	75	118
B. Bound tests								
B1. Real GDP growth is at historical average minus one standard deviations in 2017-2018	73	77	83	84	86	87	98	130
B2. Primary balance is at historical average minus one standard deviations in 2017-2018	73	76	78	76	74	72	64	57
B3. Combination of B1-B2 using one half standard deviation shocks	73	76	79	80	80	80	84	102
B4. One-time 30 percent real depreciation in 2017	73	95	93	91	89	86	80	81
B5. 10 percent of GDP increase in other debt-creating flows in 2017	73	81	79	77	75	73	65	58
PV of Debt-to-Revenue Ratio 2/								
Baseline	155	153	160	160	157	153	134	119
A. Alternative scenarios								
A1. Real GDP growth and primary balance are at historical averages	155	157	168	169	168	167	164	168
A2. Primary balance is unchanged from 2016	155	154	162	162	160	159	155	157
A3. Permanently lower GDP growth 1/	155	155	165	167	168	168	182	296
B. Bound tests								
B1. Real GDP growth is at historical average minus one standard deviations in 2017-2018	155	165	190	198	204	210	237	327
B2. Primary balance is at historical average minus one standard deviations in 2017-2018	155	163	180	180	177	174	157	144
B3. Combination of B1-B2 using one half standard deviation shocks	155	163	182	187	190	192	203	256
B4. One-time 30 percent real depreciation in 2017	155	203	213	213	211	208	195	204
B5. 10 percent of GDP increase in other debt-creating flows in 2017	155	173	182	182	179	176	159	146
Debt Service-to-Revenue Ratio 2/								
Baseline	14	22	20	26	24	24	22	20
A. Alternative scenarios								
A1. Real GDP growth and primary balance are at historical averages	14	22	20	28	25	26	26	26
A2. Primary balance is unchanged from 2016	14	22	20	26	24	24	25	25
A3. Permanently lower GDP growth 1/	14	22	20	27	25	26	28	44
B. Bound tests								
B1. Real GDP growth is at historical average minus one standard deviations in 2017-2018	14	23	22	31	31	32	35	50
B2. Primary balance is at historical average minus one standard deviations in 2017-2018	14	22	20	31	29	27	24	24
B3. Combination of B1-B2 using one half standard deviation shocks	14	23	21	31	28	30	31	39
B4. One-time 30 percent real depreciation in 2017	14	25	26	35	33	34	34	42
B5. 10 percent of GDP increase in other debt-creating flows in 2017	14	22	21	36	25	29	24	24

Sources: Country authorities; and staff estimates and projections.
1/ Assumes that real GDP growth is at baseline minus one standard deviation divided by the square root of the length of the projection period.
2/ Revenues are defined inclusive of grants.

Figure A2. Dominica: Indicators of Public Debt Under Alternative Scenarios, 2016-36 (Active Scenario) 1/



Sources: Country authorities; and staff estimates and projections.
1/ The most extreme stress test is the test that yields the highest ratio on or before 2026.
2/ Revenues are defined inclusive of grants.
3/ Customized scenarios.
4/ Suddent stop in CBI scenario assumes a complete stop of CBI inflows starting in FY2018/19.

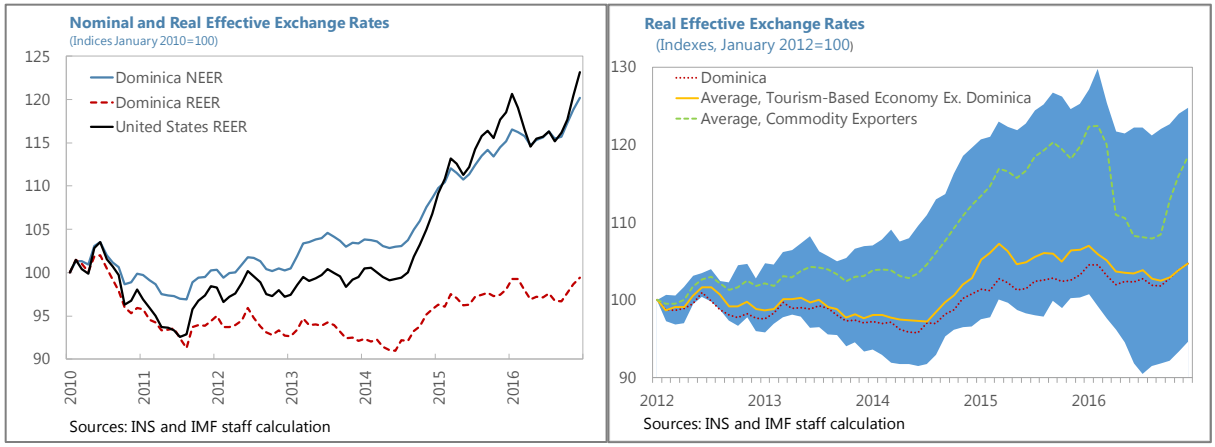
Annex II. External Sector Assessment

Dominica's external position in 2016 is assessed to be weaker than the level consistent with medium-term fundamentals and desirable policy settings. Achieving a strong and sustainable external sector requires policies and structural reforms geared at increasing external competitiveness over the medium term. Reserves are assessed to be adequate, with sufficient shock-absorbing capacity.

1. The current account deficit is expected to widen in the near term. As the collapse of manufacturing exports, the reconstruction-driven increase in imports and the decline in transfers were only partially offset by a decline in oil imports, higher-than-expected tourism receipts and citizenship-by-investment (CBI) revenues, Dominica's current account deficit is estimated to have widened from 8.0 percent of GDP in 2015 to 11.8 percent of GDP in 2016. Given the projected further increase in imports for reconstruction activity and the moderation in CBI revenues, the current account deficit is expected to widen further over the next three years before normalizing at a lower level.

A. Nominal and Real Effective Exchange Rate Assessment

2. Dominica's real effective exchange rate appreciated (REER) slightly, due to the strengthening of the US dollar. Dominica's nominal effective exchange rate (NEER) appreciated by 3.6 percent on average in 2016 mainly because of the strengthening of the U.S. dollar to which the EC dollar, the regional currency, is pegged. In comparison, the CPI based REER appreciated by 0.9 percent during the same period. Measures of the REER in tourism-competitor Caribbean countries suggest a mild deterioration of about 2.5 percent in Dominica's competitiveness in 2016. Further appreciation could continue in the near future, should relatively stronger growth and higher interest rates in the United States and uncertainty around Brexit cause the U.S. dollar to strengthen. Moreover, inflation is expected to accelerate to around 2 percent in the medium term in Dominica, providing an additional boost to the appreciation of the real exchange rate.



B. External Stability Assessments

3. The quantitative estimates of Dominica's external position suggest that it is weaker than the level consistent with medium-term fundamentals and desirable policy settings.

- Drawing on the EBA-Lite current account (CA) regression model (after accounting for Dominica-specific factors), staff assesses the cyclically-adjusted current account balance and cyclically-adjusted current account norm to be -12.2 percent of GDP and 0.7 percent of GDP, respectively in 2016. The latter is mostly driven by the significant increase in reserves in 2016 (see next section). The resulting current account gap is thus -12.9 percent of GDP. However, macroeconomic policies contributed +2.0 percent of GDP to the gap, mostly driven by fiscal policy (1.5 percent of GDP), private credit (0.3 percent of GDP) and reserve adequacy (0.2 percent of GDP). The remaining gap of -14.8 percent of GDP would need to be closed by structural policies aimed at improving competitiveness over the medium term.

- The external sustainability (ES) approach sets a medium-term current account norm at the level that would bring the net foreign assets-to-GDP ratio to zero in ten years. On this basis, the estimated norm is a deficit of 6.4 percent of GDP, compared to an underlying current account deficit of 10.3 percent of GDP.

Current Account and Real Exchange Rate Assessments
(In percent of GDP)

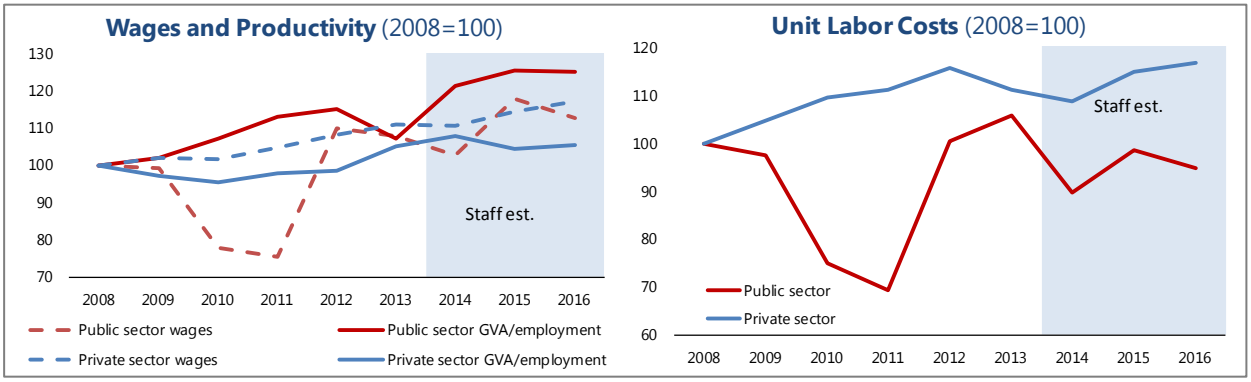
	CA Model	ES Approach	I-REER Model
CA Actual	-11.8	-11.8	
Cyclically Adjusted CA	-12.2	-10.3	
Cyclically Adjusted CA Norm	0.7	-6.4	
CA Gap	-12.9	-3.9	
Of which: Policy Gap			
Fiscal Policy	*1.5*		
Private Credit	*0.3*		
Change in Reserves	*0.2*		
Capital Controls	*0.0*		
Residual	-14.8		
Real Exchange Rate Gap[1]	50.0	15.0	5.0

Source: IMF staff estimates.

1/ Data reflect over or undervaluation (in percent). Positive numbers indicate that REER is overvalued and negative numbers indicate that the REER is undervalued.

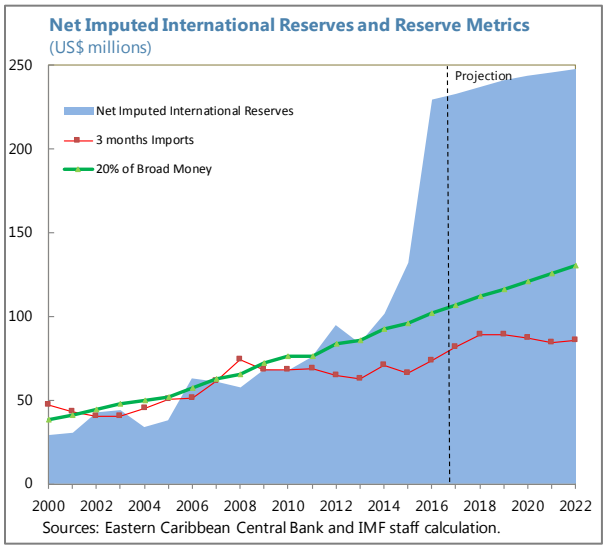
- The EBA-Lite methodologies suggest a wide range of Real Effective Exchange Rate (REER) gap estimates. Each method indicates that the REER is overvalued. Specifically, according to the CA model, a REER adjustment of 50 percent would be necessary to close the current account gap of -14.8 percent of GDP. At the same time, the ES approach and the Index REER model found that the REER needs to be adjusted by 15 and 5 percent, respectively.

4. Recent developments in wages and productivity also indicate a weakening of Dominica's external competitiveness based on unit labor costs. Since 2008, a period characterized by low growth in Dominica in the context of the global financial crisis, private sector wages have increased by 18 percent in excess of average labor productivity, leading to an increase in unit labor costs (text charts). In the public sector, cumulative wage growth was broadly aligned with labor productivity changes. Public sector compensation, however, shows significant volatility due to triennial back-payments that are typically agreed with labor unions.



C. Reserve Adequacy

5. Dominica's reserve position significantly improved in 2016. Dominica is a member of the Eastern Caribbean Currency Union. Under the quasi-currency board arrangement, foreign assets and liabilities of the Eastern Caribbean Central Bank (ECCB) cannot be directly assigned to an individual country. The imputed reserves method is thus used as a proxy for net foreign assets held with the ECCB.1 At an estimated level of US$229 million at end-2016, Dominica's imputed reserves covered 9.4 months of imports (forward cover) and 45 percent of broad money, exceeding the typical benchmarks of reserve adequacy such as 3 months of imports and 20 percent of broad money.2,3 The significant increase in the level of



[1] Imputed reserves of the individual ECCB member countries are calculated as the differences between the assets held by the member countries with the ECCB and the liabilities of the member countries to the ECCB.

[2] IMF 2015, *Assessing Reserve Adequacy –Specific Proposals*. Reserve adequacy assessments for currency unions should consider the reserve needs of the consolidated union level. This should be supplemented by a discussion of factors that have a bearing on the size of reserves, such as the union's financial architecture and supportive institutions, and the correlation of shocks faced by union members.

[3] Information on short term debt and other liabilities is unavailable and consequently assessments against other reserve adequacy metrics such as the IMF's composite Assessment Reserve Adequacy (ARA) metric cannot be computed.